

Foundations of Computer Science (26CSE0103)

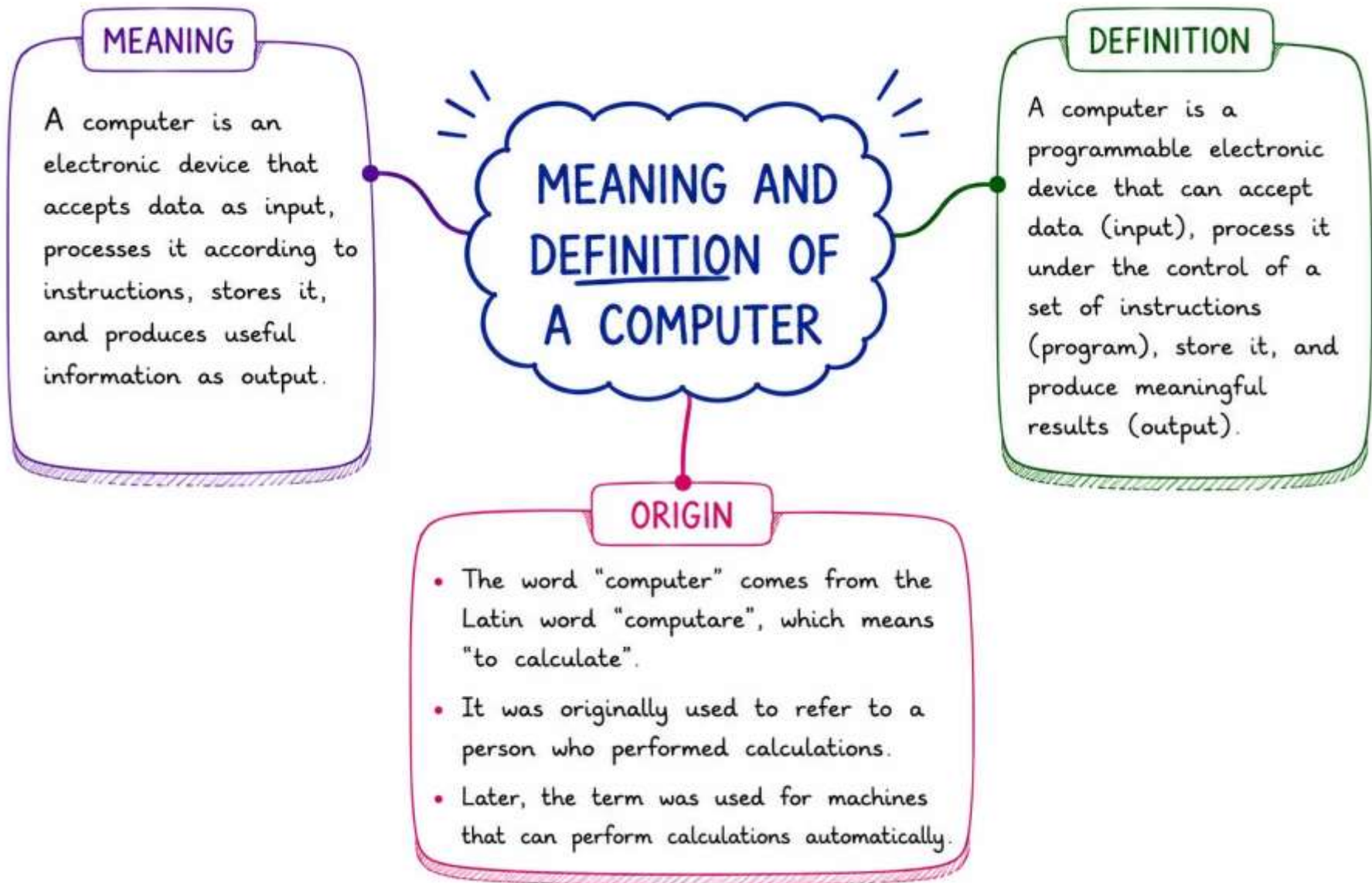
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UNIT-1 -Introduction to Computers

- Characteristics,
- Applications in daily life (Education, Healthcare, Banking, Engineering),
- Evolution and Generations of Computers (Generations I–V).
- Block Diagram of a Computer System: Functional units - input, output, memory, CPU, ALU, Control Unit, and CPU Registers, description of data bus, address bus, control bus, component interaction.

Computer?



What do computer do?



 Input, Process, Output, & Store
data

Input  **Process**  **Output**



Store Data



Definition: A computer is an electronic machine that takes data as input, processes it, and gives the result as output. It works according to instructions (programs) given by the user and performs calculations, data analysis, file management, graphic design, internet browsing, and many other tasks quickly and accurately.

The working system of a computer depends on

- **Hardware** (physical parts like monitor, keyboard, CPU)
- **Software** (programs and applications).

Hardware provides the physical infrastructure. The central processing unit (CPU), the mouse, storage and other components are examples of hardware components. The hardware components are the real components of a computer.

Software instructs the hardware on what tasks to perform. Software components of a computer are the set of instructions which we store and execute on our system, which are known as software components.

Characteristics of a computer

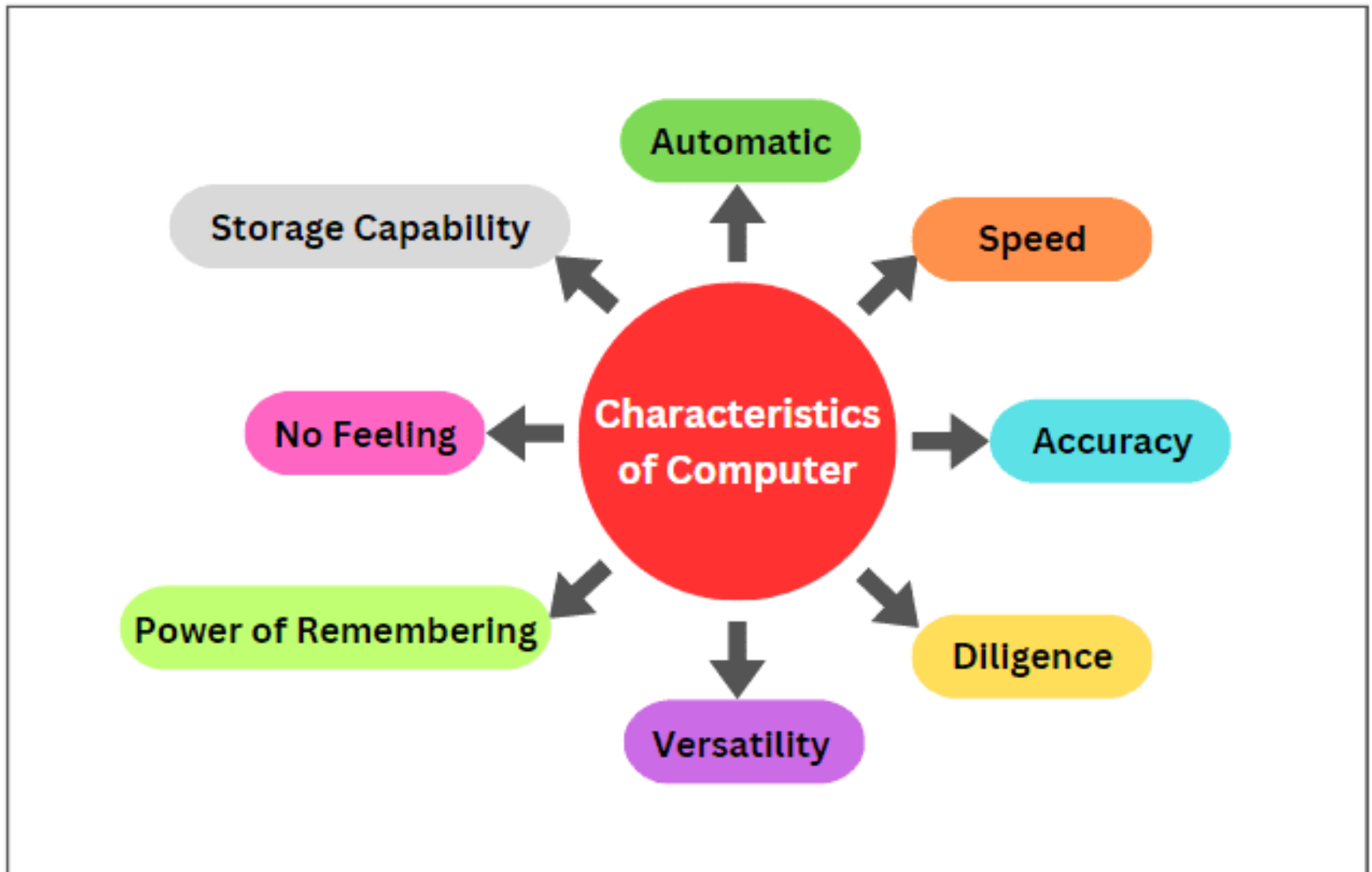


Fig.1.1 Characteristics of a Computer

Speed

A Computer can perform data processing jobs very fast, usually measured in **microseconds** (10^{-6}), **nanoseconds** (10^{-9}), and **picoseconds** (10^{-12}).

Accuracy

- Computers work free from errors.
- If the input and program are correct, there is no mistake in the output.
- It provides 100% accuracy in mathematical and logical operations.

Storage Capacity

- Computers have the ability to store large amounts of data.
- Data can be stored using hard disks, SSDs, pen drives, cloud storage, etc.
- Data is safely kept for a long time and can be accessed easily when needed.

Diligence

- Unlike human beings, a computer never gets tired and does not lose its concentration.
- It can continuously work with a great accuracy for hours, creating no error and without rumbling.
- **e.g.** if hundred million calculations have to be performed, a computer machine will perform the last one with exactly the same accuracy and speed as it did the first one.

Automatic

Given a job, a computer can work on it automatically without human intervention.

Versatility

- It is capable of performing multitasks at a time simultaneously.
- For example, we can use it to play music, as well as write a letter in the MS Office simultaneously.

Power of Remembering

- Computers have the power of storing any amount of data or information.
- We can store any information for several years, and recalled when we need it. It forgets or loses certain information only when it is asked to do so.

No Feeling

- A computer is an electronic device that does not have any feelings, emotions, taste, knowledge and experience.
- It does not get exhausted even after long hours of work.

Storage Capability

- A computer has huge storage capability to store the information for a long time.
- It has two types of storages: primary storage and secondary storage.
- *Primary storage* is the main memory of a computer that stores the data temporarily. While *secondary* storage is a permanent storage for storing data.

1. Communication

Email & Messaging: Gmail, Outlook, WhatsApp, Slack.

Video Conferencing: Zoom, Google Meet, Microsoft Teams.

Social Media: Facebook, Instagram, X (Twitter), LinkedIn.

2. Education & Learning

Online Classes & E-Learning Platforms: Coursera, Udemy, Swayam-NPTEL.

Research & Study: Google Scholar, Wikipedia, digital libraries, IEEE expolre

Interactive Learning Tools: Simulations, virtual labs, and educational games.

3. Business & Work

Office Productivity: Word processors, spreadsheets, presentations.

Data Management: Databases, CRM systems (Salesforce, HubSpot).

Automation: Payroll systems, inventory control, AI chatbots.

4. Banking & Finance

Online Banking: Checking balances, transferring money, paying bills.

Digital Payments: PayPal, Google Pay, Apple Pay.

Stock Trading: Platforms like Robinhood, E-TRADE.

5. Healthcare

Medical Records Management: Electronic Health Records (EHR).

Diagnostics & Imaging: MRI, CT scans, X-rays.

Telemedicine: Online doctor consultations and health monitoring devices.

6. Entertainment

Streaming: Netflix, YouTube, Spotify.

Gaming: PC games, VR experiences.

Digital Art & Music Production: Photoshop, FL Studio.

7. Shopping

E-commerce: Amazon, eBay, local online marketplaces.

Price Comparison Tools: Finding best deals online.

Order Tracking & Delivery Services.

8. Transport & Navigation

Ride-hailing Apps: Uber, Lyft.

GPS Navigation: Google Maps

Traffic & Public Transport Updates.

9. Home Automation

Smart Devices: Alexa, Google Home, smart thermostats.

Security: CCTV monitoring, smart locks.

Appliance Control: Smart ovens, washing machines.

10. Creative Work

Design & Media Production: Video editing, 3D modeling, animation.

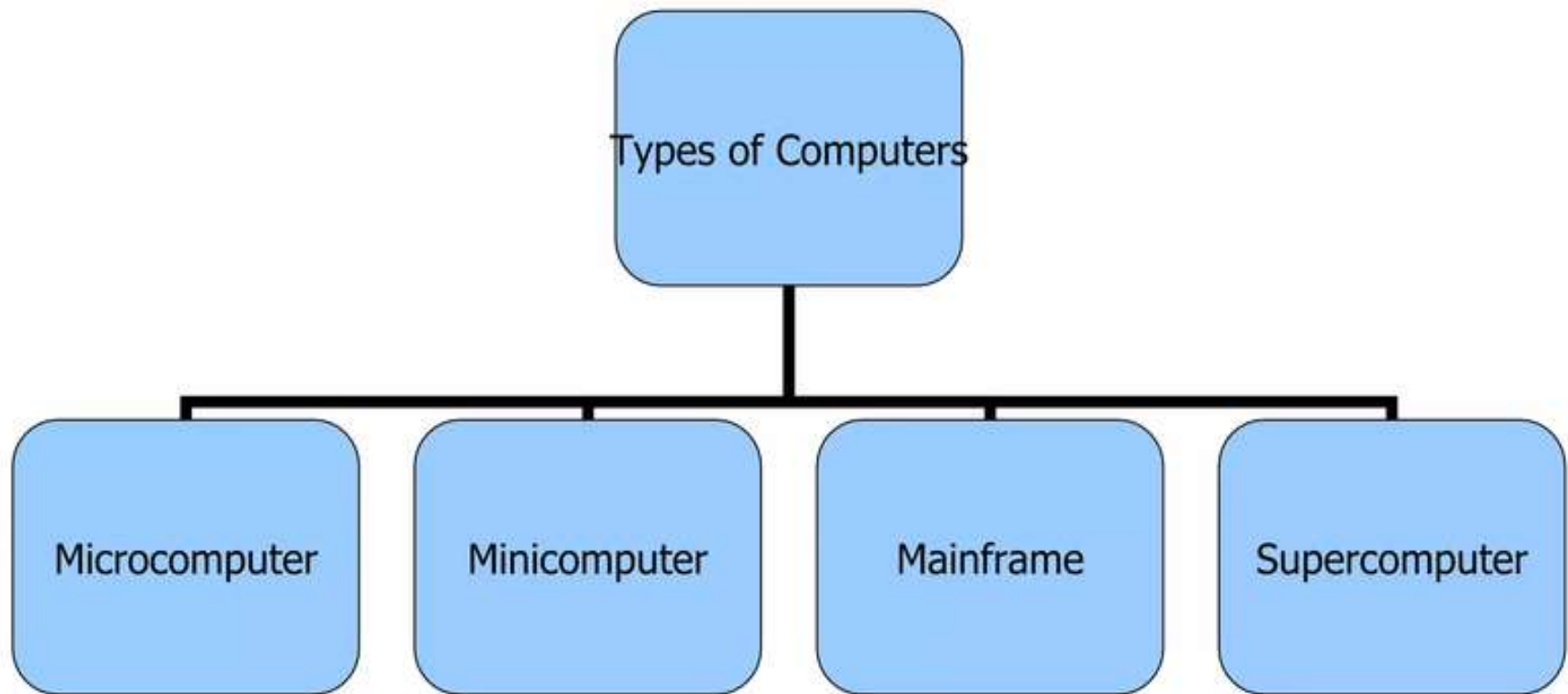
Writing & Publishing: Blogs, books, newsletters.

Photography & Image Editing.

- A computer plays an important role in the medical field also.
- For example in the Intensive Care Unit (ICU) in the hospital, the computer keeps track of everything going inside the patient's body such as blood pressure and heartbeat, etc.
- Keeping a record of patients and medicines also becomes easy through the use of the computer.
- **Diagnostic System** — The computer is used to collect data and identify the cause of disease and illness.
- **Lab-diagnostic System** — All tests can be done and the reports are prepared by computer.
- Patient Monitoring System — Computers are also used to check the patient's signs for abnormality such as CT Scan, MRI, ECG, etc.
- **Pharmacy Information System** — Computers are used to check medicine labels, expiry dates, harmful side effects, etc. *Software:* Epic Willow Pharmacy, Oracle Health Pharmacy.
- **Surgery** — Computers are used in performing surgery. *Software:* Mako Smart Robotics

Computers are grouped by size, power, and how they process data. The main categories include supercomputers, mainframes, minicomputers, and microcomputers (personal computers).

Types of Computers





1. Personal Computer (Microcomputer)



2. Minicomputer



3. Server



4. Workstation



5. Mainframe



6. Supercomputer

Supercomputer: The fastest and most powerful machines. They handle massive scientific calculations, climate research, and weather forecasting

Mainframe Computer: Large, highly secure systems built to process millions of transactions simultaneously. Used by banks and big companies

Minicomputer (Mid-range): Medium-sized computers that sit between mainframes and microcomputers. They support multiple users for business tasks.

Microcomputer (Personal Computer): Small, affordable devices for individual use. This group includes desktops, laptops, tablets, and smartphones.

Microcomputer Models



Desktop



Laptop



Notebook



Subnotebook



Palmtop



Analog Computer: Analog computers are used mainly for measurement, monitoring, simulation, and control of continuously changing physical quantities. Aircraft and flight simulation

Digital Computer: Works with discrete values or binary data (0s and 1s). This is the standard type used for modern laptops, phones, and servers.

Hybrid Computer: Combines features of both analog and digital types. It uses analog speed with digital memory, often found in medical and industrial setups.



Feature	Microcomputer (PC)	Minicomputer	Mainframe Computer	Supercomputer	Workstation	Server
Definition	Small computer for a single user	Mid-range computer for multiple users	Large computer for high-volume data processing	Fastest computer for scientific and engineering computations	High-performance computer for professional applications	Computer that provides services and resources to other computers over a network
Memory (RAM)	4–64 GB	16–256 GB	Hundreds of GB to several TB	Several TB to PB	16–512 GB	32 GB to several TB
Storage Capacity	256 GB–4 TB	Several TB	Hundreds of TB to PB	PB or more	1–20 TB	Several TB to PB
Operating System	Windows, Linux, macOS	UNIX, Linux	z/OS, Linux	Linux-based HPC OS	Windows, Linux, UNIX	Windows Server, Linux, UNIX
Primary Use	Personal use, education, office work, internet browsing, gaming	Used by small/medium organizations for database management	Banking, railway/airline reservations, insurance, government records,	Weather forecasting, climate research, space research, nuclear research, scientific simulations, AI	Engineering design (CAD), 3D graphics, animation, scientific research, software development, video editing	Hosting websites and applications, file sharing, database management, email services, cloud services,
Approximate Cost	₹30,000–₹2 lakh	₹5–50 lakh	₹2–20 lakh	₹5 lakh–₹5 crore	₹10–500 crore	₹500 crore–₹5,000+ crore

- Blaise Pascal invented the first *mechanical adding machine* in 1642
- Baron Gottfried Wilhelm von Leibniz invented the first *calculator for multiplication* in 1671
- *Keyboard machines* originated in the United States around 1880
- Around 1880, Herman Hollerith came up with the concept of *punched cards* that were extensively used as input media until the late 1970s
- **Charles Babbage** is considered to be the father of the modern digital computer
 - He designed the “Difference Engine” in 1822
 - He designed a fully automatic analytical engine in 1842 for performing basic arithmetic functions
 - His efforts established several principles that are fundamental to the design of any digital computer

- “*Generation*” in computer talk is a step in technology. It provides a framework for the growth of computer industry
- Originally it was used to distinguish between various hardware technologies, but now it has been extended to include both hardware and software
- Till today, there are five computer generations

Generation & Description

- **First Generation**
The period of first generation:
1946-1959.
Vacuum tube based.
- **Second Generation**
The period of second
generation: 1959-1965.
Transistor based.
- **Third Generation**
The period of third generation:
1965-1971.
Integrated-Circuit based.
- **Fourth Generation**
The period of fourth
generation: 1971-1980.
VLSI microprocessor based.
- **Fifth Generation**
The period of fifth generation:
1980-onwards.
ULSI microprocessor-based

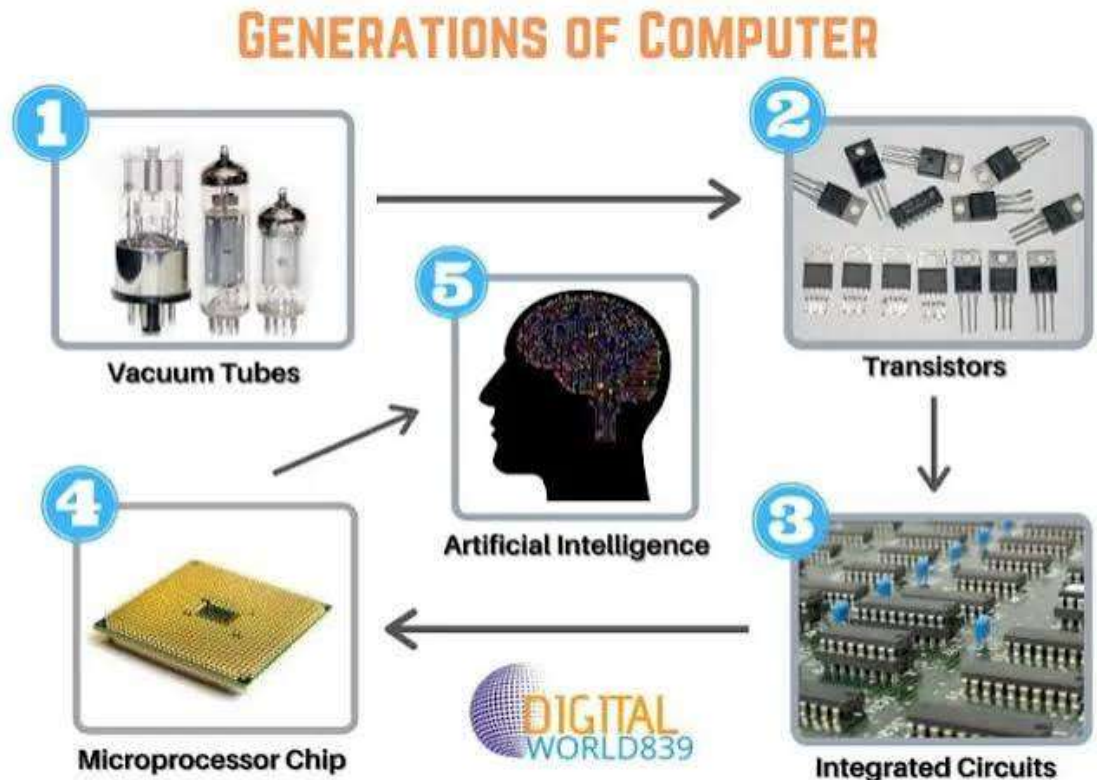


Fig. 1.2 Key technologies behind five generations of a computer

1. First Generation (1940s–1956s)



Technology: Vacuum tubes.

Language: Machine language (binary).

Size: Huge (room-sized).

Examples: Electronic Numerical Integrator and Computer (ENIAC),
Universal Automatic Computer I (UNIVAC I).

Features: Very slow, produced a lot of heat, high electricity use,
need of AC, Non-portable.



2. Second Generation (1956s–1963s)



Technology: Transistors (replaced vacuum tubes).

Language: Assembly language.

Size: Smaller than first generation.

Examples: IBM 1401, CDC 1604.

Features: Faster, more reliable, less heat, lower power consumption, still very costly, AC required.

3. Third Generation (1964s–1971s)

Technology: Integrated Circuits (ICs) — many transistors on a single chip.

Language: High-level languages (COBOL, FORTRAN).

Examples: IBM 360 series, PDP-8.

Features: Cheaper, smaller, more efficient, started to be used for business and government.



4. Fourth Generation (1970s–Present)

Technology: Microprocessors (all computer functions on a single chip bringing an entire CPU onto a single chip).

Language: Advanced high-level languages (C, C++).

Examples: Apple II, IBM PC.

Features: Personal computers become common, rapid growth in networking.

5. Fifth Generation (Present & Future)

Technology: Artificial Intelligence (AI), Machine Learning and parallel processing.

Language: Natural language processing, AI programming languages (Python, Prolog).

Examples: AI assistants (Siri, ChatGPT), supercomputers.

Features: High speed, massive data processing, focus on learning and decision-making,

- ULSI technology,
- More user-friendly interfaces with multimedia features,
- Availability of very powerful and compact computers at cheaper rates

Difference between Evolution and Generation

Aspect	Evolution of Computers	Generation of Computers
Meaning	Overall development of computers from earliest tools to modern AI systems.	Specific stage in computer history defined by a major change in core hardware technology.
Scope	Broad — includes all technological, software, and usage changes over time.	Narrow — focuses mainly on hardware improvements in each stage.
Focus	Continuous progress and trends.	Distinct phases separated by technological breakthroughs.
Time Span	Covers the entire history of computing.	Covers only the duration of a particular generation (e.g., 1950s–1960s for 2nd gen).
Example	From abacus → mechanical calculators → ENIAC → modern AI.	1st Gen: Vacuum tubes, 2nd Gen: Transistors, 3rd Gen: Integrated circuits, etc.
Nature	Story of <i>how</i> and <i>why</i> computers improved.	Classification system based on <i>what</i> technology powered the computers.

Advantages of Computers

Speed and Efficiency: Computers process large amounts of data quickly, automating tasks and saving time.

Accuracy: Computers perform calculations and tasks with high precision, reducing human error.

Storage Capacity: They can store vast amounts of data in a compact space, easily accessible and organized.

Multitasking: Can handle multiple tasks simultaneously, improving productivity

Versatility: Support a wide range of applications, from education to entertainment and business.

Disadvantages of Computers

Health Issues: Prolonged use can lead to eye strain, back pain, and repetitive strain injuries.

Cost: Initial setup, maintenance, and software updates can be expensive.

Dependency: Over-reliance on computers can reduce critical thinking and manual skills

Security Risks: Vulnerable to hacking, viruses, and data breaches if not properly secured

Technical Issues: Hardware or software failures can disrupt work and require technical expertise to fix.

Von Neumann Architecture

From an architectural point of view, there are the two basic models: Von Neumann and Harvard architectures. These models explain the **organization of memory** and **processing units** in a computer system.

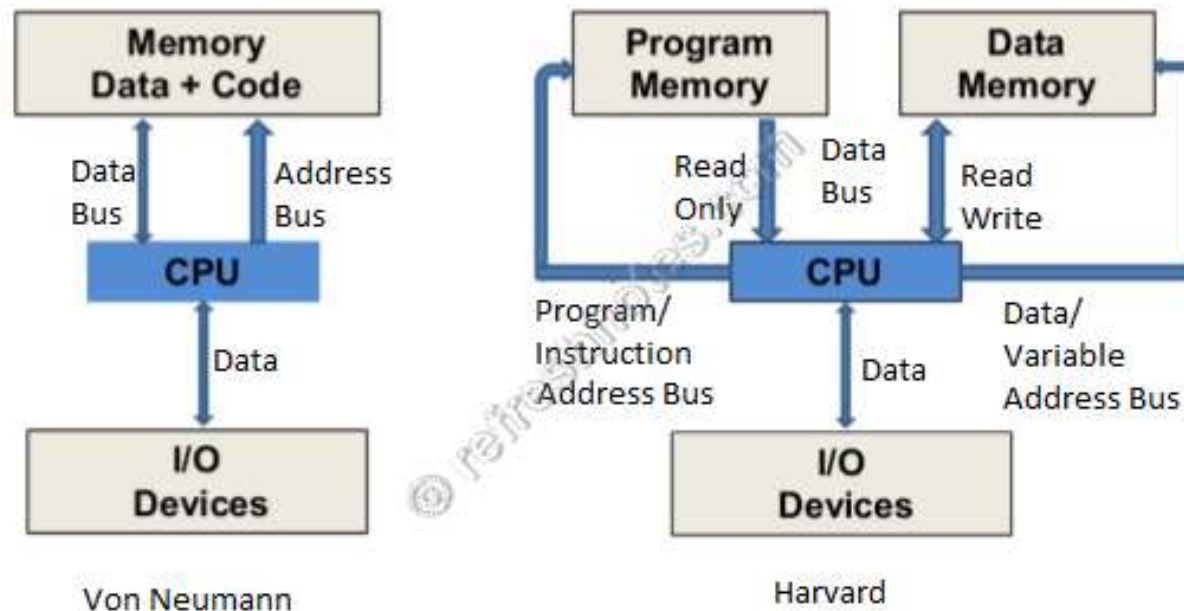
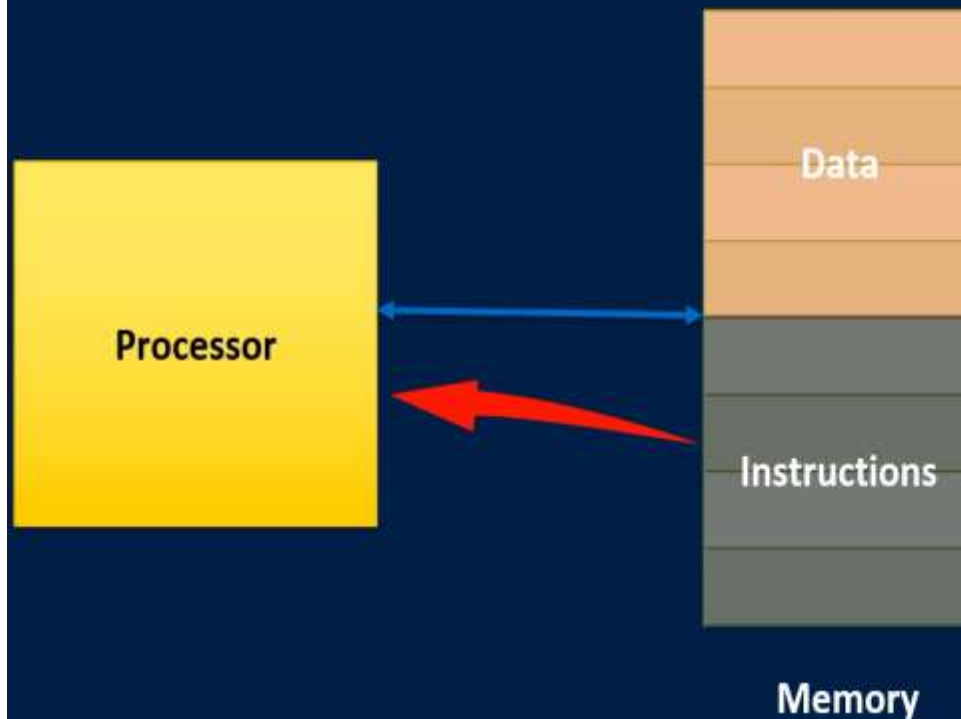


Fig. 1.7 Von Neumann Vs Harvard architecture

- Historically, there have been 2 types of Computers:
 - **Fixed Program Computers** - Their function is very specific, and they couldn't be reprogrammed, e.g., calculators.
 - **Stored Program Computers** - These can be programmed to carry out many different tasks, applications are stored on them, hence the name.

Stored Program Architecture



Data and instructions are stored in the memory.

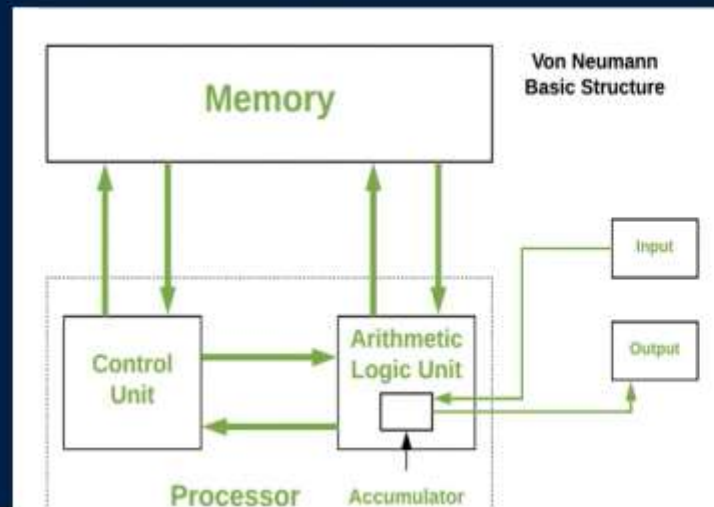
The processor fetches the instructions and executes them one after the other.

Von Neumann Architecture

Von Neumann Architecture was given by John Von Neumann in the year 1945. It is also known as Princeton Architecture.

It is a type of stored program architecture in which data and instructions are stored in the main memory.

The machine uses the same set of control signals for both data and instruction fetches.

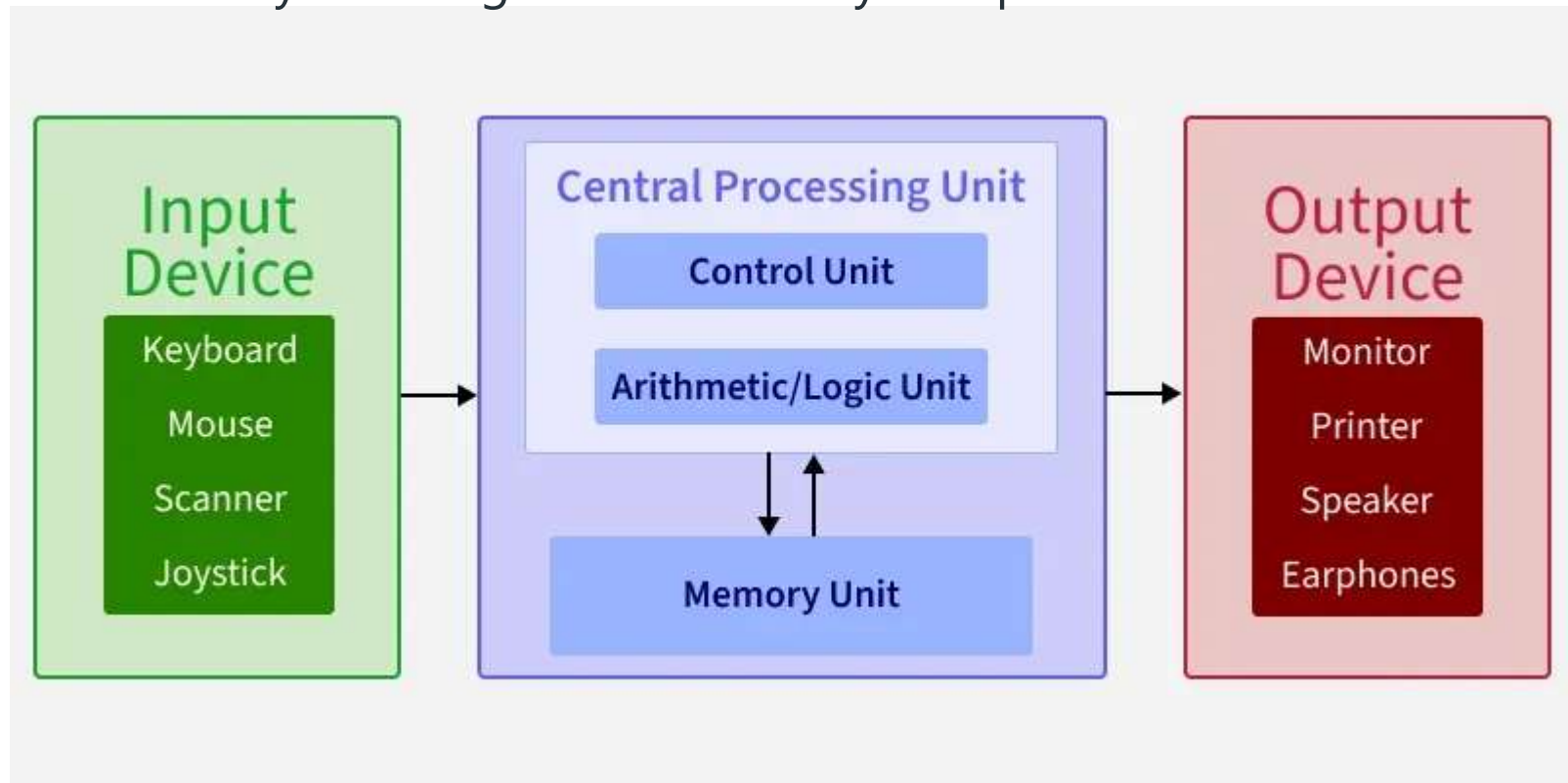


- **Stored Program Concept** (programs and data in the same memory).
- **Sequential execution** (one instruction after another).
- **Single memory** for both data and instructions.
- **Control unit** directs the flow of execution.

Basic components of a computer system



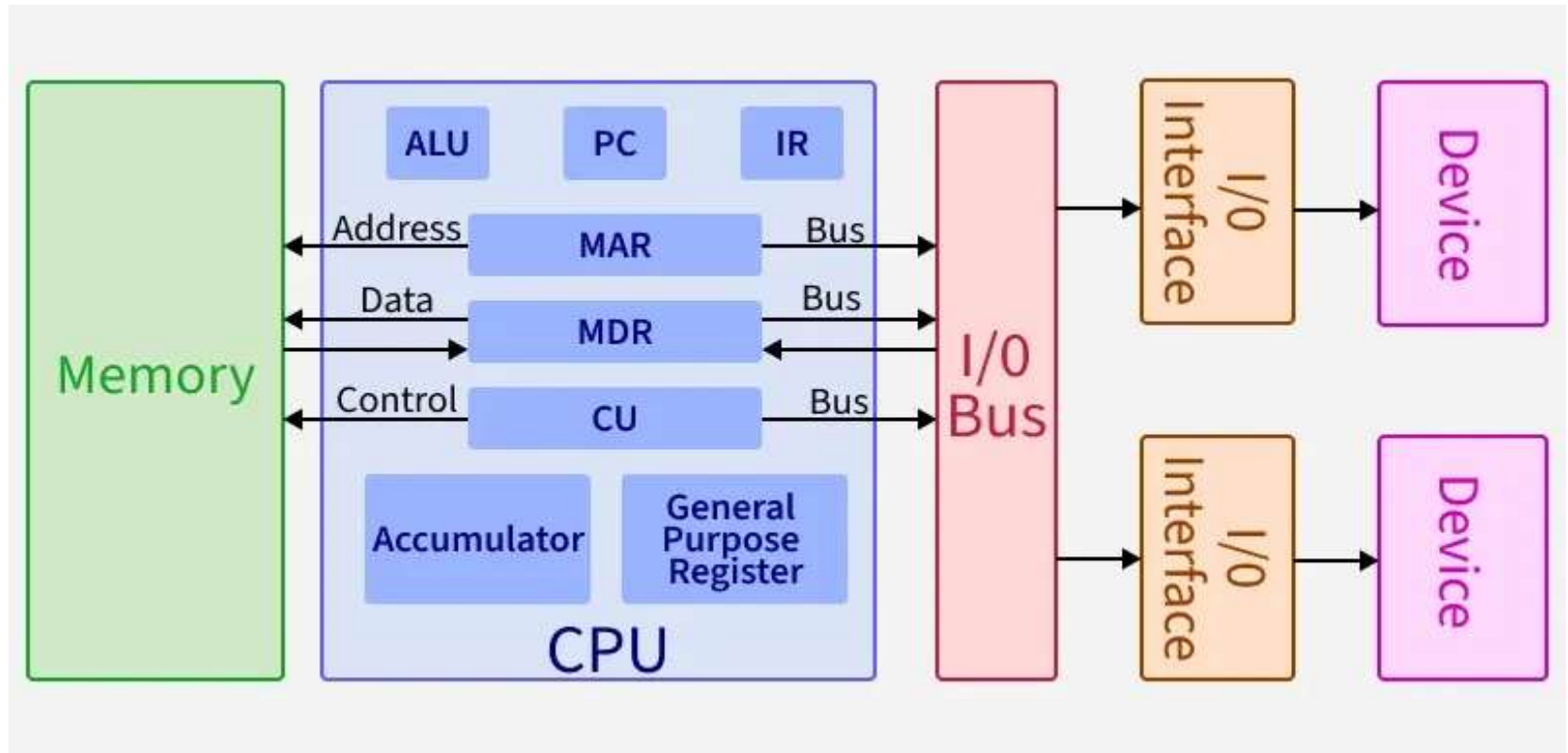
Particularly focusing on the memory and processor



- Memory:** This is where data and instructions are stored. It is a crucial part of the computer system that allows for the storage and retrieval of information.
- Input:** This refers to the devices or methods through which data is entered into the computer system.
- Output:** This refers to the devices or methods through which data is presented to the user or other systems.
- Processor:** The processor, or CPU, is the central component that carries out the instructions of a computer program. It includes the ALU and Control Unit.
- Control Unit:** This component manages the operations of the computer. It directs the flow of data between the CPU and other components.
- Arithmetic Logic Unit (ALU):** The ALU performs arithmetic and logical operations. It is responsible for calculations and decision-making processes.

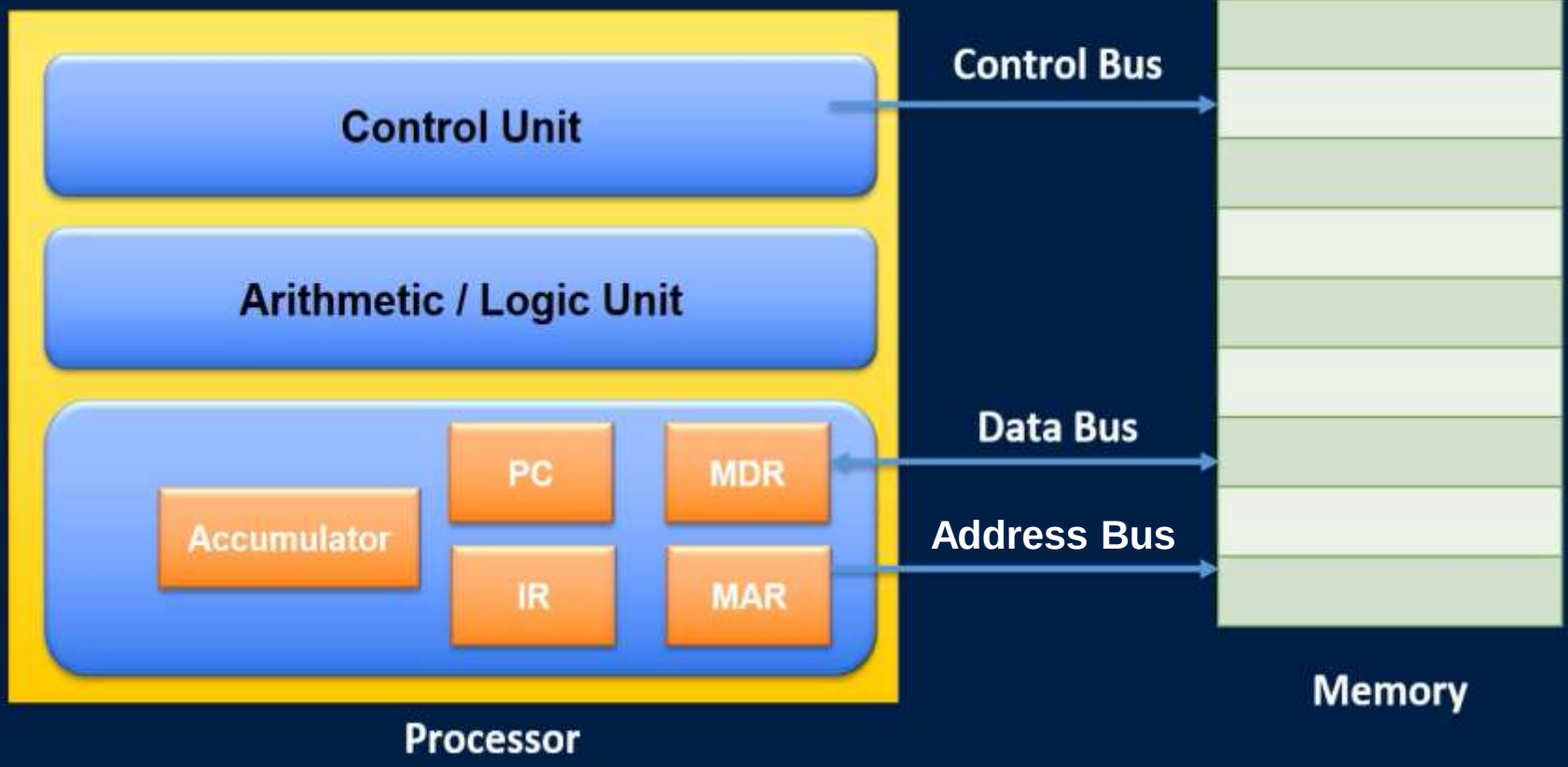
Basic CPU structure, illustrating ALU

In this architecture, both data and instructions are stored in the same memory and share a common bus for communication





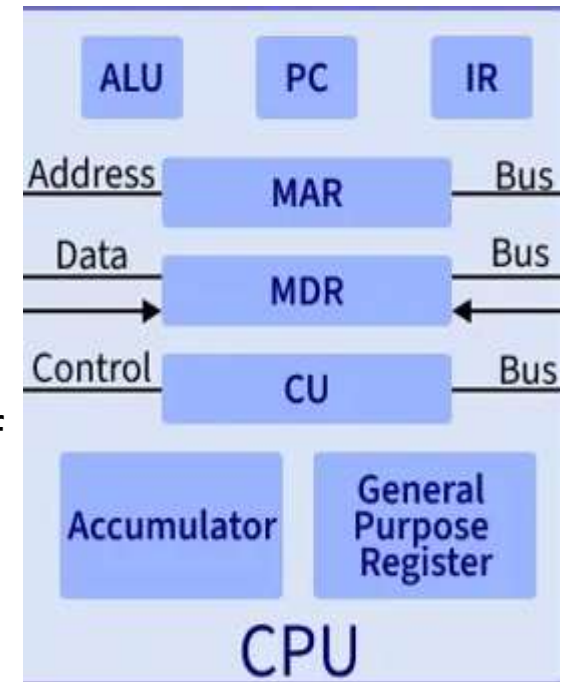
Bus



CPU (Central Processing Unit)

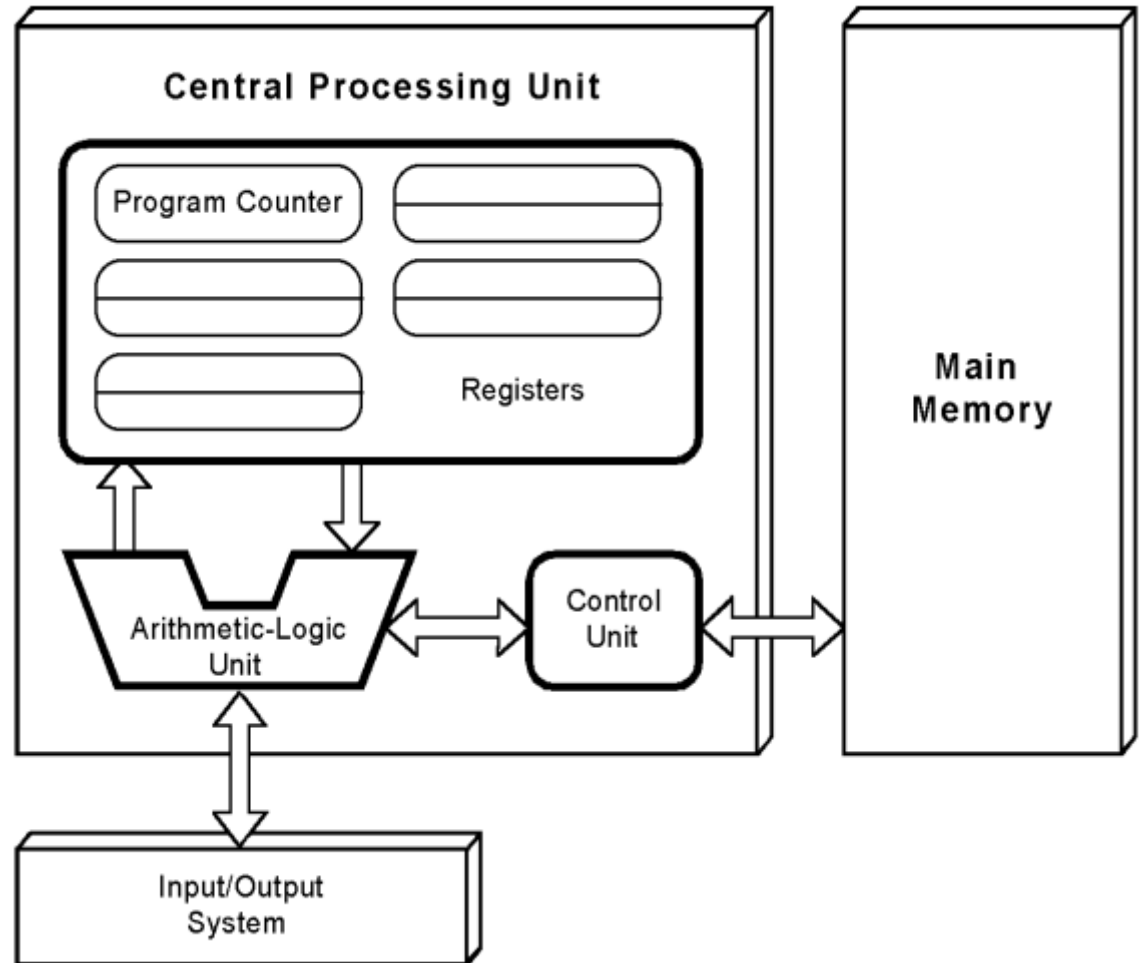
The CPU is the core processing unit that executes instructions. It consists of:

- **ALU (Arithmetic Logic Unit):** Performs arithmetic and logical operations (e.g., addition, subtraction, comparisons).
- **PC (Program Counter):** Keeps track of the address of the next instruction to be executed.
- **IR (Instruction Register):** Holds the current instruction being executed.
- **MAR (Memory Address Register):** Stores the address of the memory location being accessed.
- **MDR (Memory Data Register):** Temporarily holds data being transferred to or from memory.
- **CU (Control Unit):** Coordinates the activities of the CPU, managing the flow of data and instructions.
- **Accumulator:** A register that stores intermediate results of arithmetic and logic operations.
- **General Purpose Registers:** Used for temporary storage of data during processing.



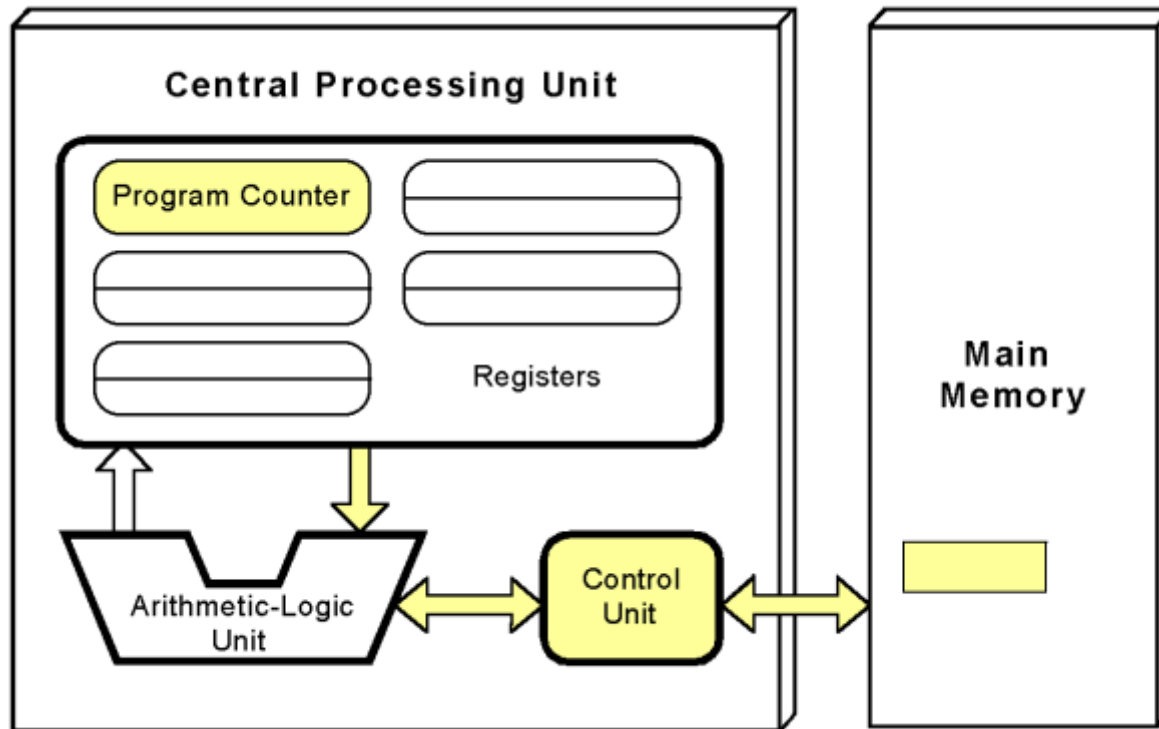
The von Neumann Model

- This is a general depiction of a von Neumann system:
- These computers employ a fetch-decode-execute cycle to run programs as follows . . .



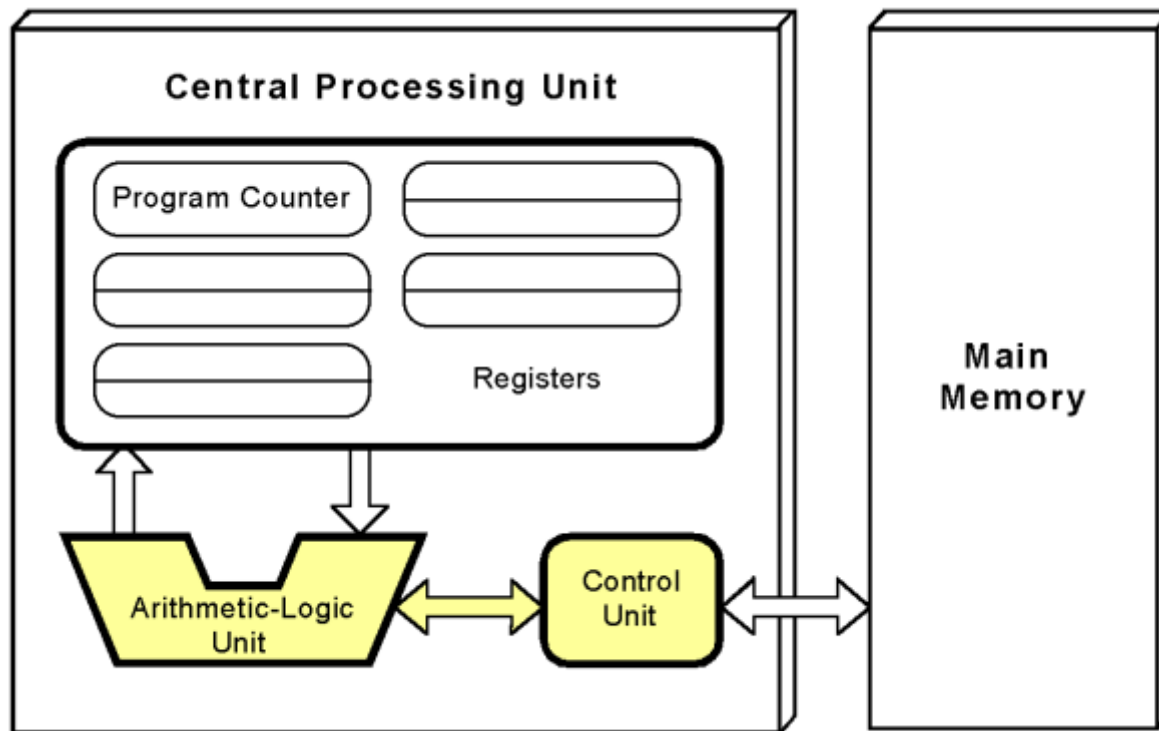
The von Neumann Model

The control unit fetches the next instruction from memory using the program counter to determine where the instruction is located.



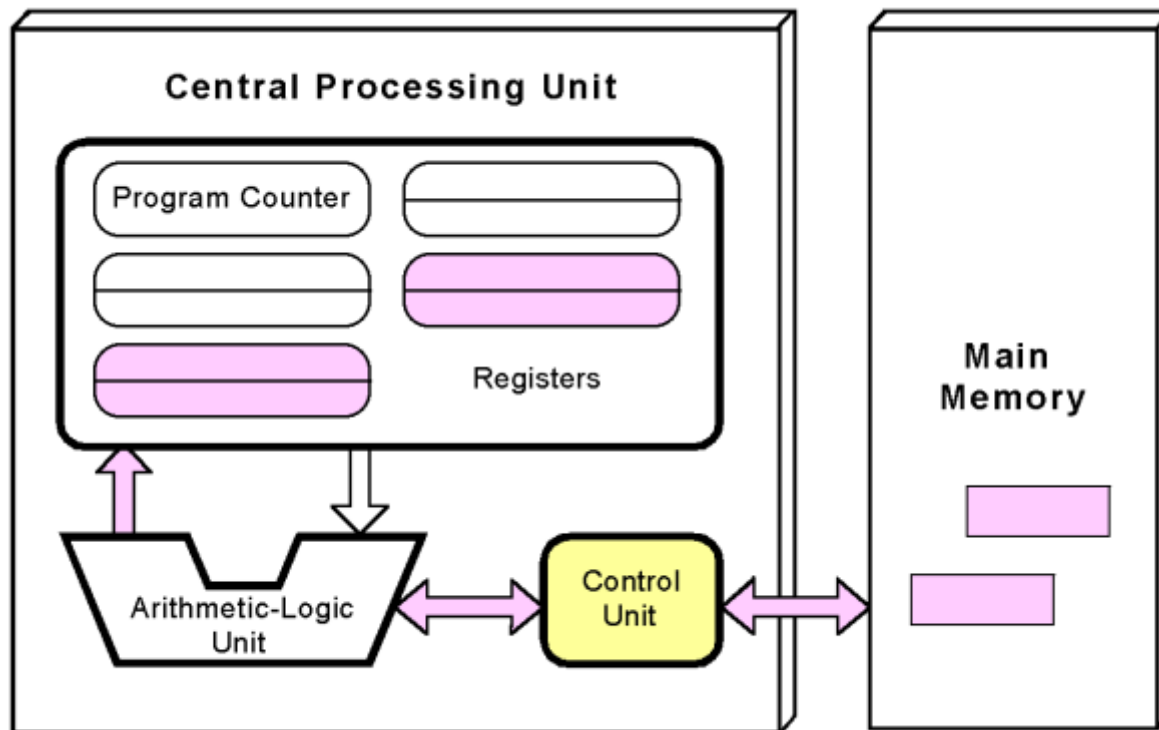
The von Neumann Model

The instruction is decoded into a language that the ALU can understand.



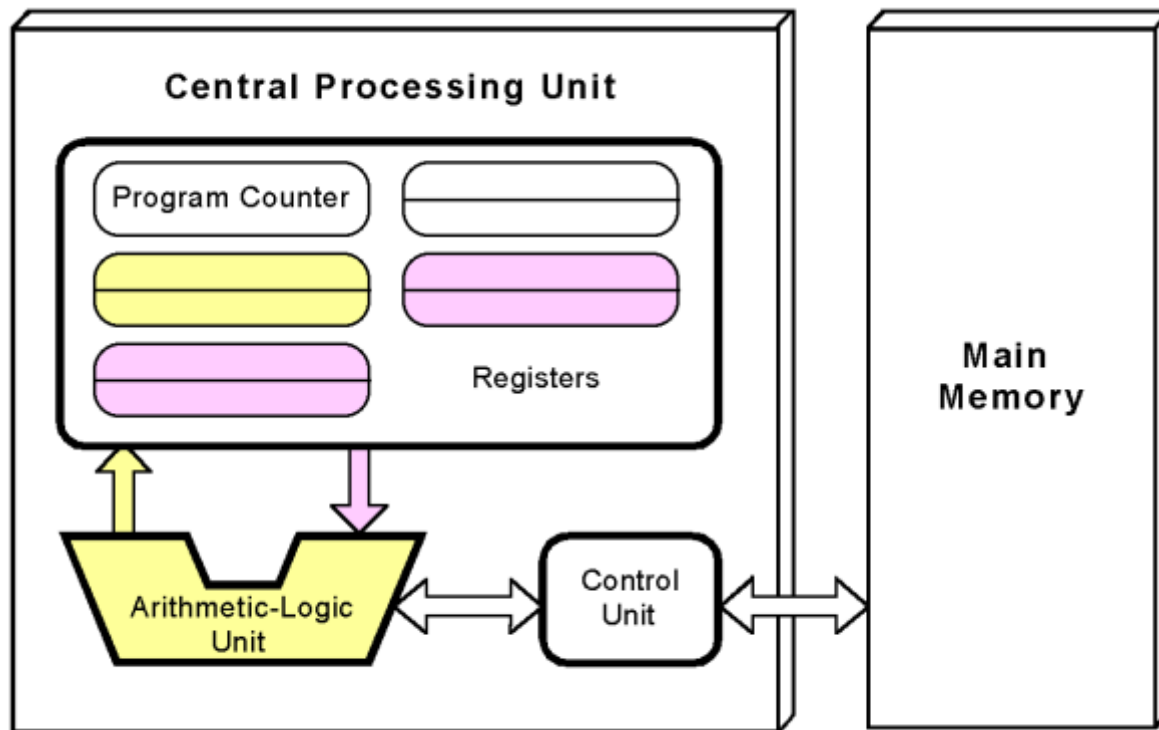
The von Neumann Model

Any data operands required to execute the instruction are fetched from memory and placed into registers within the CPU.



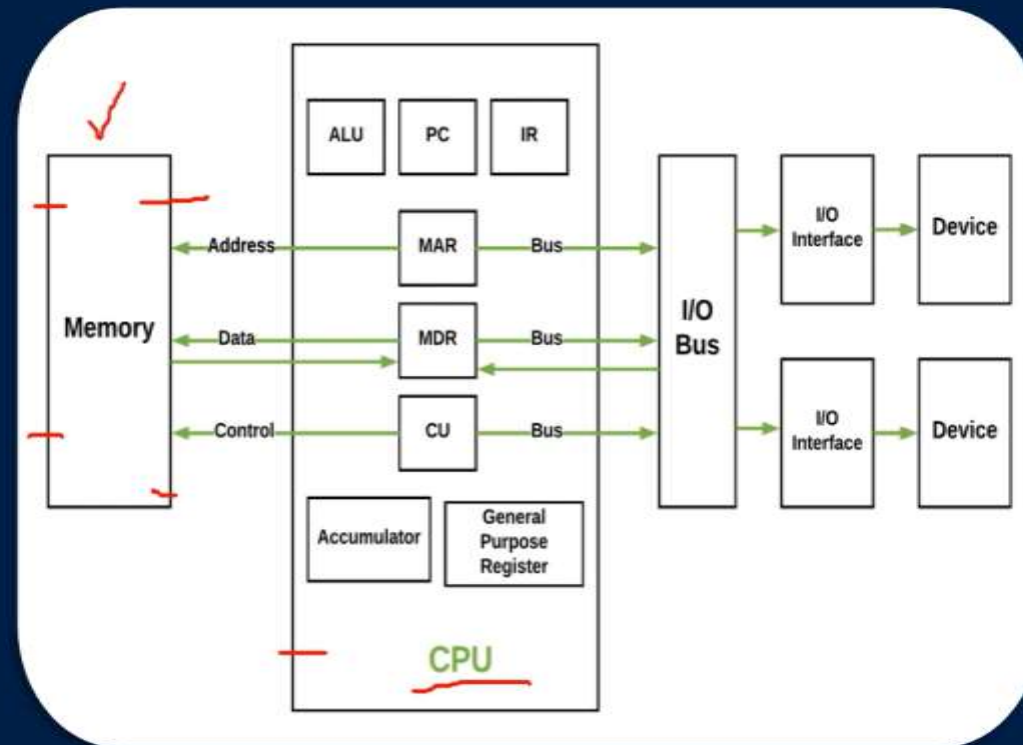
The von Neumann Model

The ALU executes the instruction and places results in registers or memory.



1. **Fetch:** Instruction is fetched from memory using the Program Counter (PC).
2. **Decode:** The Control Unit interprets (decodes) the instruction.
3. **Execute:** ALU or other parts execute the instruction.
4. **Repeat:** The PC updates to the next instruction.

Von Neumann Bottleneck



The shared bus system between the processor and the main memory does not permit both data as well as instructions to be fetched simultaneously causing a limited throughput and thus is a limitation.

Advantages of Von Neumann Architecture

- **Simplified Design:** Uses a single memory for data and instructions, reducing hardware complexity.
- **Cost-Effective:** Lower production costs due to fewer components.
- **Flexibility:** Can run various programs and makes it suitable for general-purpose computing.
- **Ease of Programming:** Unified memory structure simplifies software development.
- **Widely Adopted:** Forms the foundation of most modern computers hence, ensures widespread compatibility.

Limitations of Von Neumann Architecture

- **Single bus system** → instructions and data can't be fetched at the same time → slows down performance.
- **Memory Bottleneck:** Shared memory slows down data and instruction transfer. (Memory latency issues → CPU often waits for data)
- **Sequential Processing:** Cannot process data and instructions simultaneously. (→ not suitable for parallel processing)
- **Scalability Issues:** Struggles with high-performance tasks requiring rapid memory access.
- **Energy Inefficiency:** Frequent memory access increases power consumption.
- **Latency:** Data and instruction fetch delays reduce overall system efficiency.

Applications of Von Neumann Architecture

- **General-Purpose Computing:** Powers desktops, laptops, and smartphones.
- **Software Development:** Shapes programming tools and languages due to its unified structure.
- **Education:** A foundational concept in computer science courses.
- **Gaming and Multimedia:** Supports complex applications like video games and editing software.



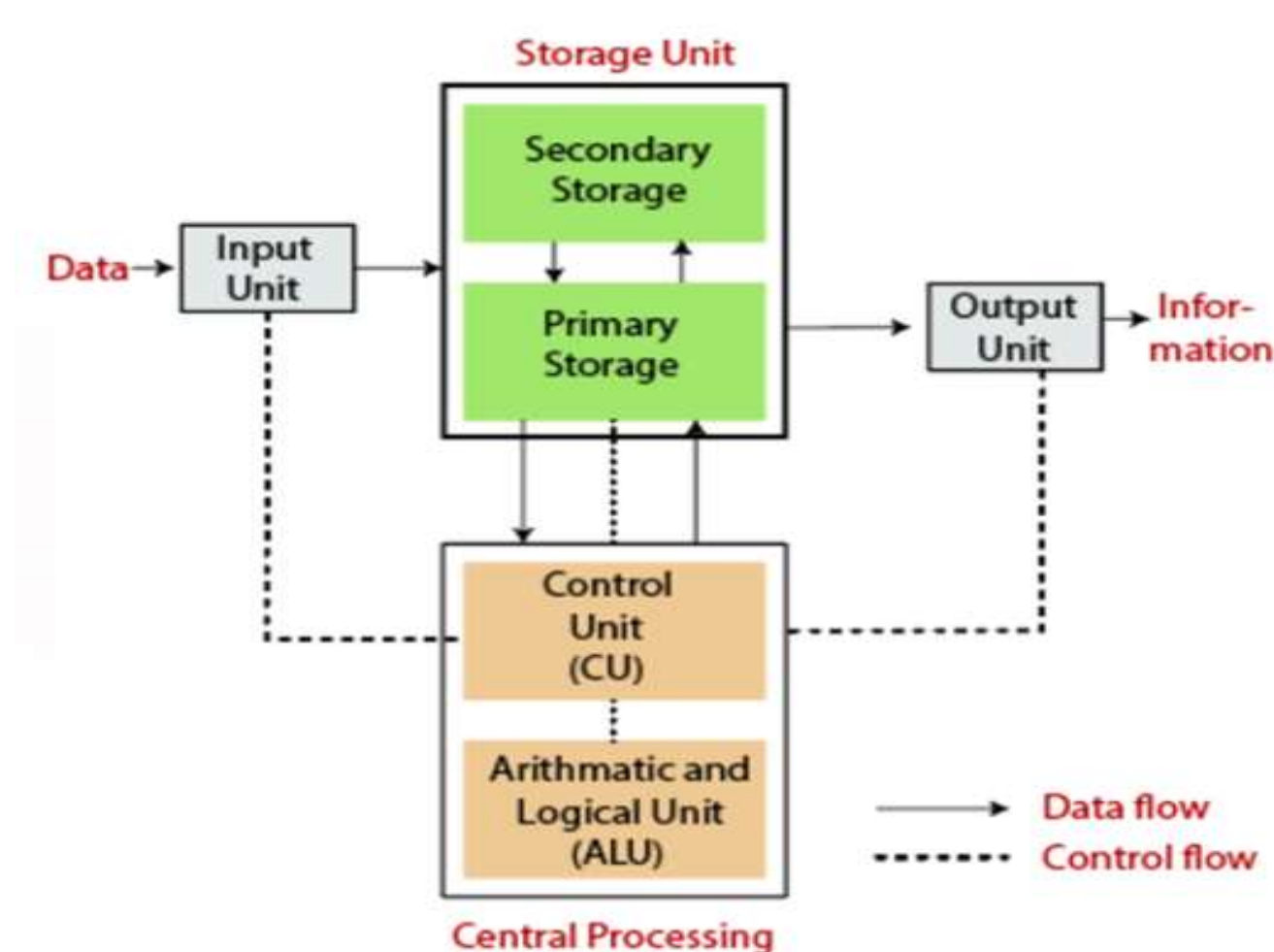
- A digital computer is defined as a device that operates upon information/data.
- To be able to process data, the computer is made of various functional units to perform its specified task.



Block Diagram of Computer System



- Block diagram provides the basic structure of a computer system where the main parts of the computer system are represented.

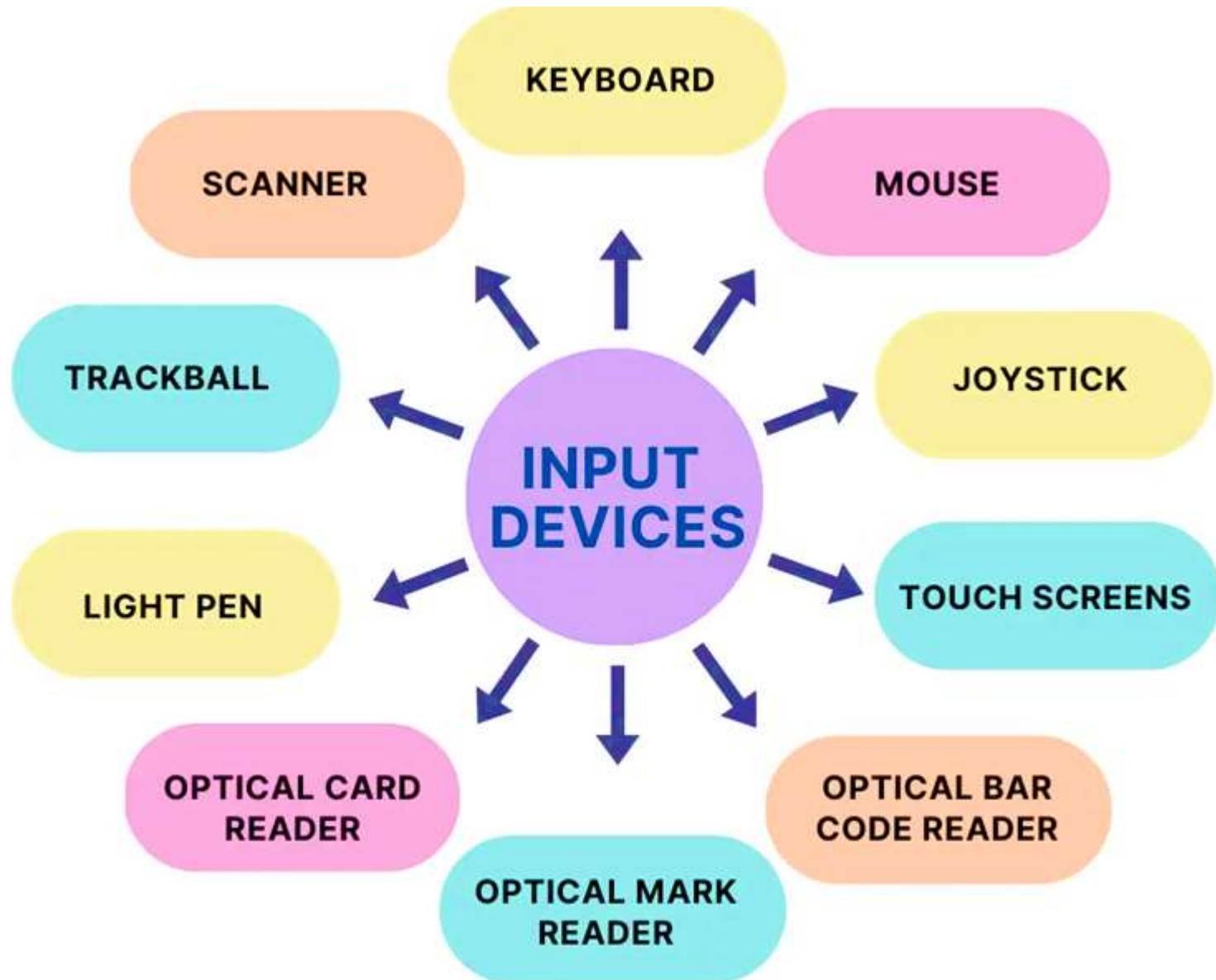


- Each block of the diagram performs particular tasks, and all blocks are well coordinated to run the system.
- The Major Components of the block diagram of the computer system are:
 - ❑ **Input Unit**
 - ❑ **Output Unit**
 - ❑ **Memory Unit**
 - ❑ **Central Processing Unit (CPU)**
 - **Control Unit (CU)**
 - **Arithmetic And Logic Unit (ALU)**

- The computer requires the input unit to receive data and information and then use it to solve problems.
- The user provides the computer with data and information.
- The input unit is made up of a variety of input devices. Some input devices are a keyboard, mouse, microphone, scanner, etc.

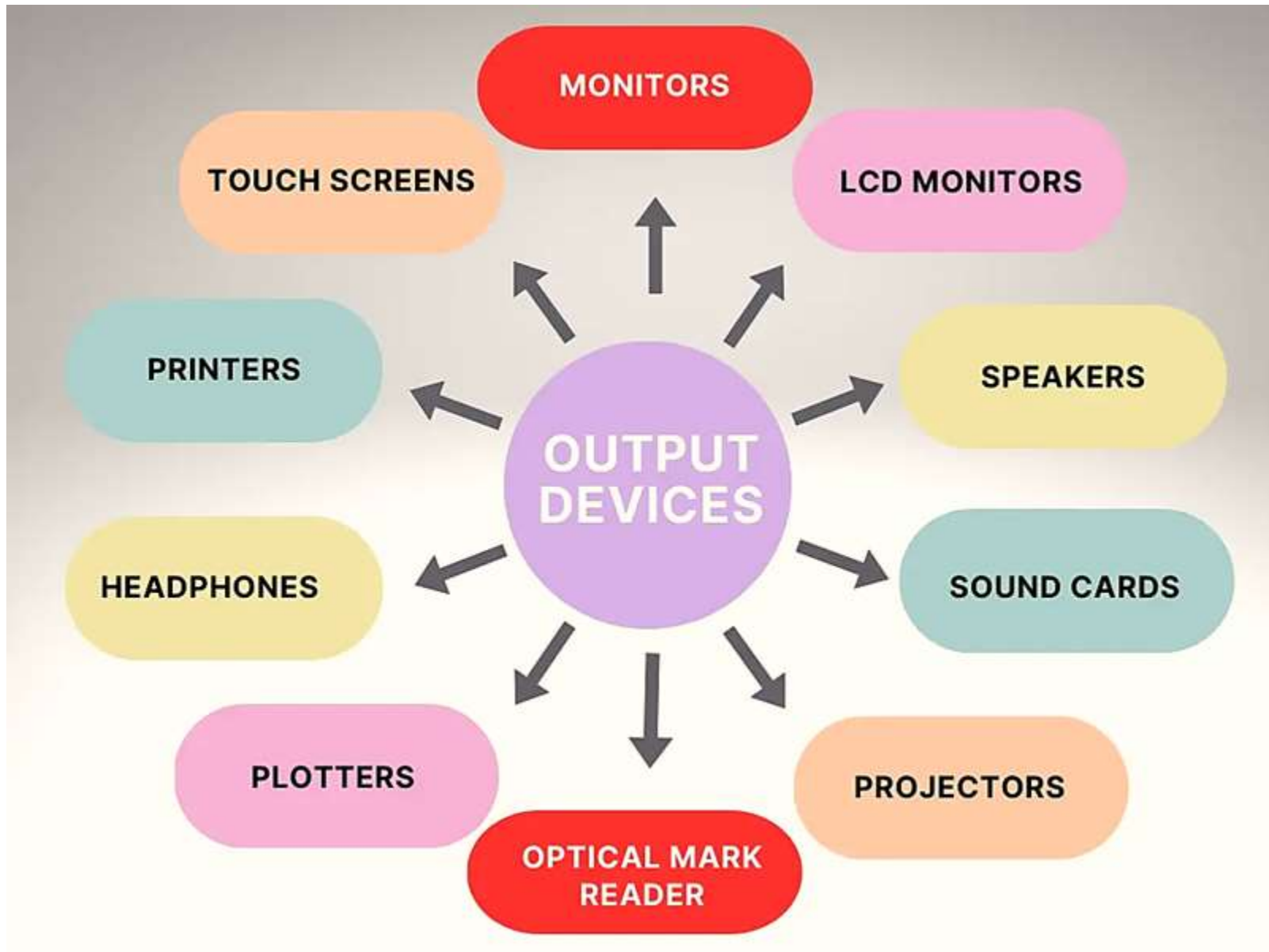
- Accept data from the user.
- Convert the data from electrical signals to computer-readable machine language.
- Instruct the CPU to receive data from the input devices.
- Provide the transformed data to the CPU through the memory unit for further processing.

Input Unit (Devices)

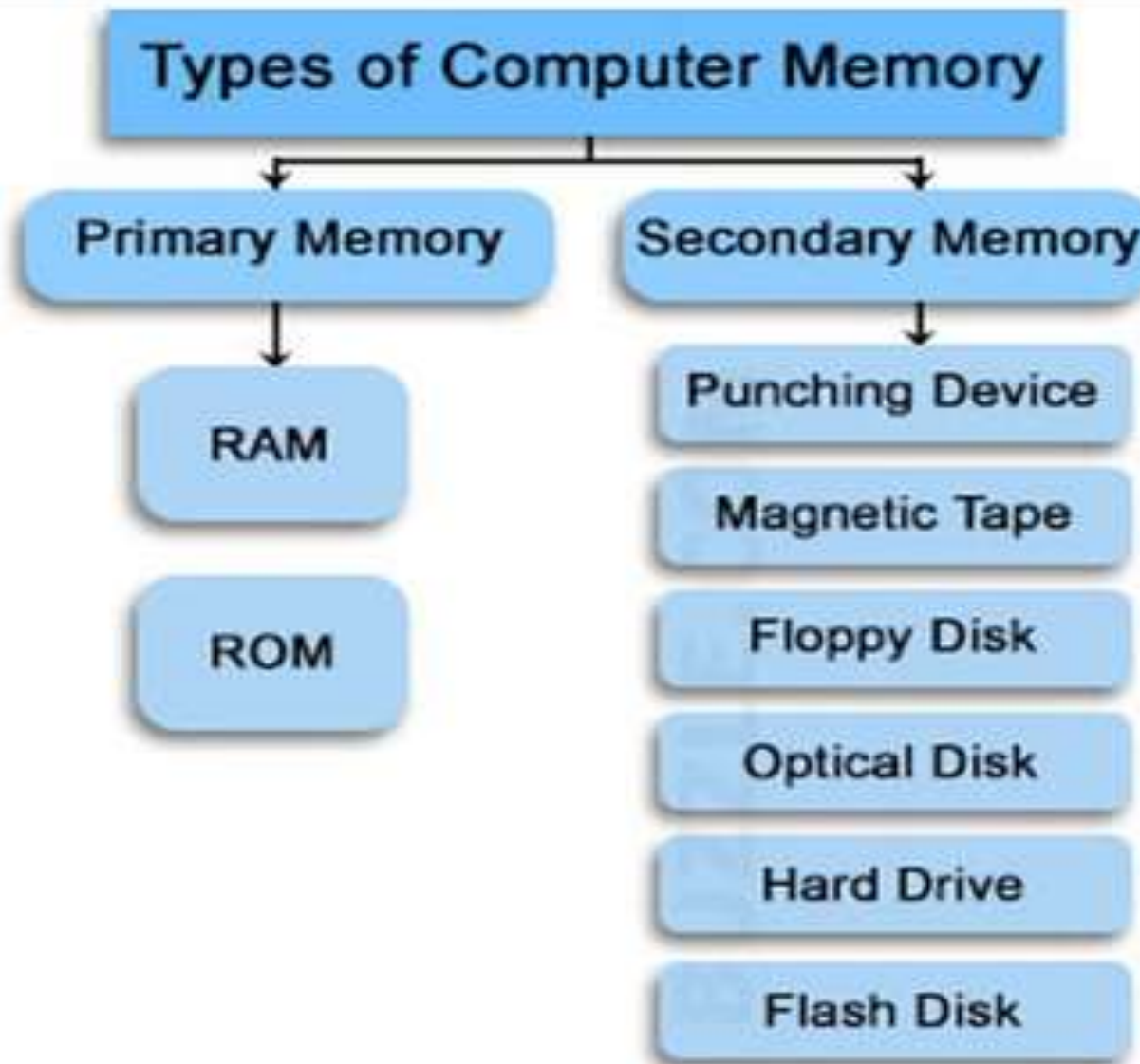


- The computer's output unit communicates the results of processing and information to the outside world.
- In general, the output unit reverses the process of the input unit, converting machine language into electronic signals readable by the output devices.
- The Visual Display Unit (VDU), usually known as the monitor, is the most commonly utilized output device.

Output Unit (Devices)



- The storage unit of the computer holds data and instructions that are entered through the input unit, before they are processed.
- It preserves the intermediate and final results before these are sent to the output devices.
- It also saves the data for the later use.



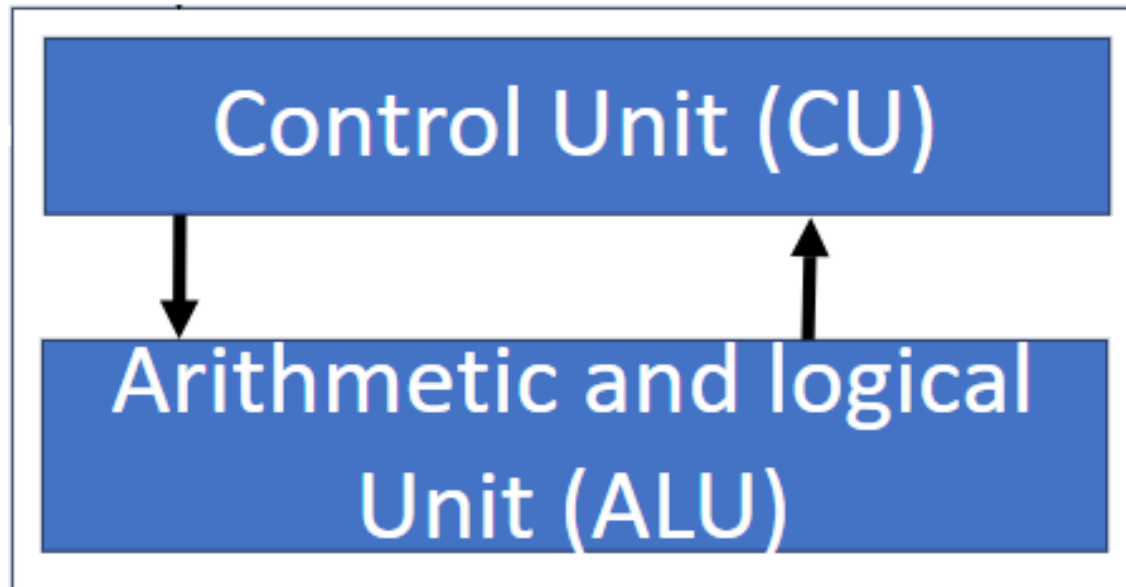
- Data stored in this memory is temporary -**volatile**.
- Typically, smaller in **size** compared to secondary memory.
- Holds currently executing programs, Intermediate data during computation.
- CPU can directly **access** primary memory.
- **Faster** as compared to secondary storage.

- Used to store data and programs **permanently**.
- It is **non-volatile**.
- Has much larger storage capacity than primary memory
- **Slower** than primary memory in terms of data access speed.
- The **cost** is less compared to the primary storage.
- Some secondary storage devices are **portable**.

- The Central Processing Unit or CPU is known as the **brain** of the computer.
- Just like the human brain controls all human activities, the CPU also takes care of all the tasks.
- The CPU is responsible for performing all the **arithmetic and logical operations** within the computer.

➤ The CPU comprises two parts:

- ❑ **ALU (Arithmetic Logic Unit)**
- ❑ **CU (Control Unit)**



- Generally referred as the **central nervous** system of the computer (directs and coordinates all the operations).
- It receives information from the main memory of computer.
- It tells the Logic Unit, Memory Unit, as well as Input or Output devices how to handle the instructions in proper order.
- It **does not process or store data**; it only controls the flow of data.

- Comprises of two terms: **Arithmetic** and **Logic**.
- Performs operations like addition, subtraction, multiplication, and division.
- Handles **comparisons** like greater than, less than, equal to, etc.
- **Doesn't** control or manage the process flow.
- Executes operations as instructed by the **Control Unit**.

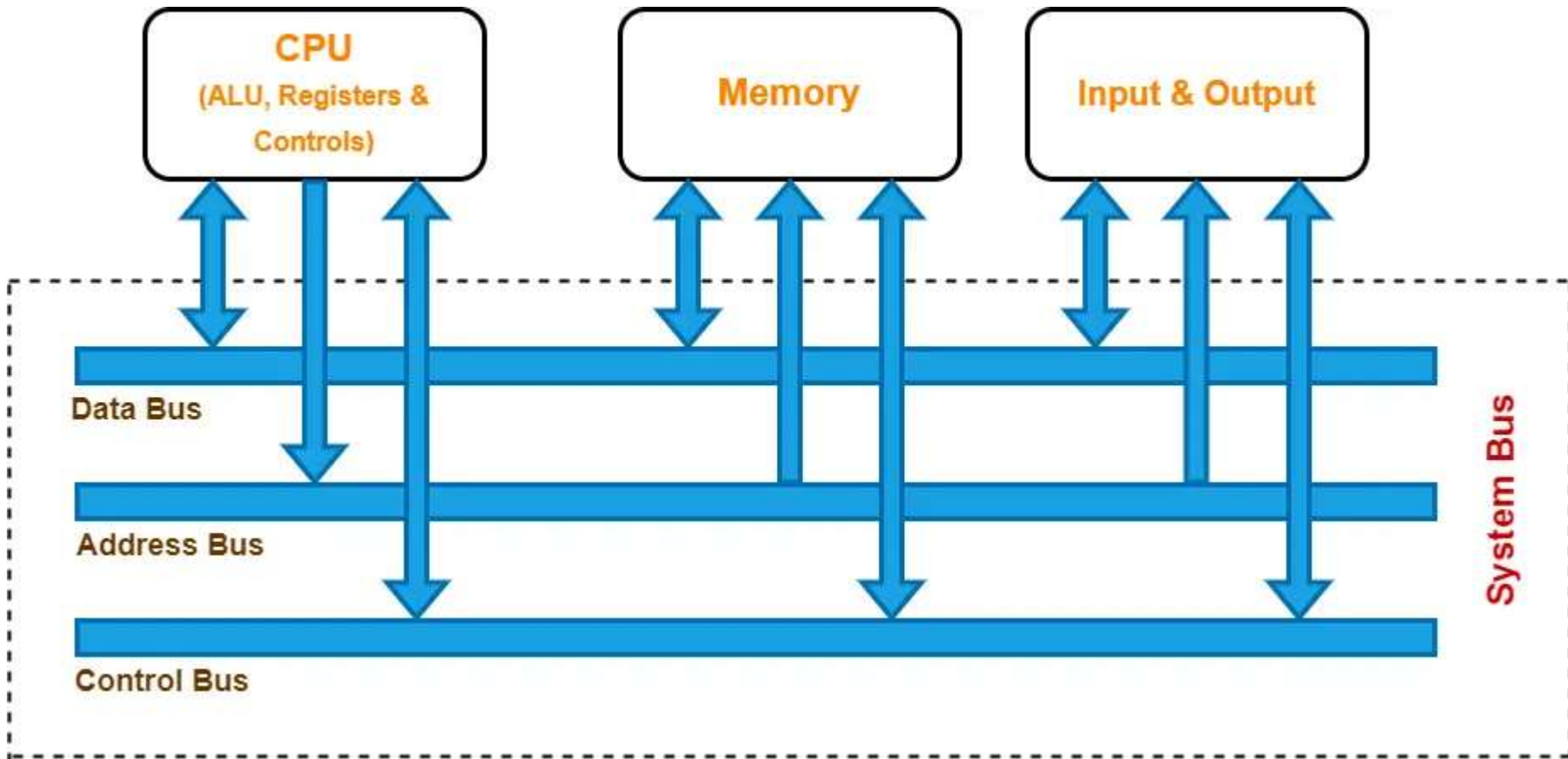
Introduction to Computer Bus



- The Computer Bus is a **communication link** used in a computer system to send **data, addresses, control signals**, and **power** to various hardware components in a computer system.
- These buses act as a **shared communication channel**.



Introduction to Computer Bus



- Dedicated for the transmitting the data between the CPU, peripheral devices and other hardware components.
- A data bus has many different features, but one of the most important feature is the **bus width**.
- Width of a data bus refers to the **number of bits** (wires) that the bus **can carry at a time**.
- The Common data bus widths include **8 bit, 16 bit, 32 bit and 64 bit**.
- The **data bus is bidirectional** because the data can flow in either direction from CPU to memory(or input/output device) or from memory to the CPU.
- The size (width) of bus determines how much data can be transmitted at one time.

Example:

- A 16-bit bus can transmit 16 bits of data at a time.
- A 32-bit bus can transmit 32 bits at a time.

- Control bus is used to **transfer the control signals** from one component to another component.
- Used by the CPU to **communicate** the devices that are connected to the computer system.
- These devices are connected with the help of cables and **printed circuits board** such **motherboard**.
- It is **control bus** assist the CPU in **synchronizing the control signals**.
- The **Control bus is bidirectional** because the data can flow in either direction from CPU to memory(or input/output device) or from memory to the CPU.
- it also transmits response signals from the hardware.

Example

- This bus is used to indicate whether the CPU is reading from memory or writing to memory.

- The computer program consists of several program instructions.
- These instructions direct the CPU to perform the desired operation.
- The CPU executes the program instructions one by one by fetching the program instructions from the main memory.
- To operate, the CPU sends either a read or write control signal on the control bus and the address of the memory location along the "Address Bus" from where the operation is to be performed.
- The **address bus is unidirectional**.
- The **size of address bus determines how many unique memory locations can be addressed**.

Example:

A system with 4-bit address bus can address $2^4 = 16$ Bytes of memory.

A system with 16-bit address bus can address $2^{16} = 64$ KB of memory

A system with 20-bit address bus can address $2^{20} = 1$ MB of memory.

Bus Type	Direction	Purpose	Key Role
Address Bus	Unidirectional	Carries memory addresses	Identifies where data should go
Data Bus	Bidirectional	Carries actual data	Moves data between components
Control Bus	Bidirectional	Carries control and sync signals	Coordinates CPU and device actions

1. Which of the following is a characteristic of a computer?

- A. Slow processing
- B. High accuracy
- C. Limited storage
- D. Emotional intelligence

Answer: B

2. Which computer application is commonly used in education?

- A. Online learning
- B. ATM withdrawal
- C. X-ray scanning
- D. CAD designing

Answer: A

3. Which technology is commonly used in healthcare?

- A. Medical imaging
- B. Word processing
- C. Spreadsheet formatting
- D. File compression

Answer: A

4. The first generation of computers primarily used:

- A. Transistors
- B. Integrated Circuits
- C. Vacuum tubes
- D. Microprocessors

Answer: C

5. The second generation of computers used:

- A. Vacuum tubes
- B. Transistors
- C. Microprocessors
- D. Artificial intelligence

Answer: B

6. Integrated Circuits were primarily associated with which generation?

- A. First
- B. Second
- C. Third
- D. Fifth

Answer: C

7. The fourth generation of computers is characterized by:

- A. Vacuum tubes
- B. Transistors
- C. Microprocessors
- D. Magnetic drums

Answer: C

8. The fifth generation is mainly associated with:

- A. Artificial Intelligence
- B. Vacuum tubes
- C. Punch cards
- D. Magnetic drums

Answer: A

9. Which unit is responsible for performing arithmetic and logical operations?

- A. Control Unit
- B. ALU
- C. Input Unit
- D. Output Unit

Answer: B

10. Which unit controls and coordinates all operations of a computer?

- A. ALU
- B. Memory Unit
- C. Control Unit
- D. Output Unit

Answer: C

Scenario 1: Online Education

A college conducts online classes using computers. Students use a keyboard and webcam to provide information, the CPU processes instructions, memory stores data temporarily, and the monitor displays the result.

1. Which functional unit receives information from the keyboard and webcam?

- A. Output Unit
- B. Input Unit
- C. ALU
- D. Control Unit

Answer: B

2. Which component performs arithmetic and logical operations on the student's data?

- A. ALU
- B. Monitor
- C. Input Unit
- D. Register only

Answer: A

3. Which unit coordinates the activities of the CPU?

- A. ALU
- B. Control Unit
- C. Output Unit
- D. Memory Unit

Answer: B

4. The monitor is an example of:

- A. Input device
- B. Processing device
- C. Output device
- D. Storage device

Answer: C

5. Which bus would carry the actual data being transferred between CPU and memory?

- A. Address bus
- B. Data bus
- C. Control bus
- D. Power bus

Answer: B

Scenario 2: Hospital

A hospital uses computers to maintain patient records, process medical information and display diagnostic reports. Doctors enter patient information and retrieve stored records when required.

1. Patient information entered using a keyboard is initially handled by the:

- A. Input Unit
- B. Output Unit
- C. ALU
- D. Printer

Answer: A

2. A diagnostic report displayed on a screen is an example of:

- A. Input
- B. Processing
- C. Output
- D. Addressing

Answer: C

3. If the computer performs calculations on a patient's test results, which CPU component performs the calculations?

- A. Control Unit
- B. ALU
- C. Register
- D. Address Bus

Answer: B

4. Which component stores instructions and data required during processing?

- A. Memory Unit
- B. Monitor
- C. Keyboard
- D. Printer

Answer: A

5. Which CPU component directs other components to perform their required operations?

- A. ALU
- B. Control Unit
- C. Register
- D. Output Unit

Answer: B

Scenario 1: Basic Linux File Management

A student is working in a Linux terminal and wants to manage files and directories.

1. Which command displays the files and directories in the current directory?

- A. pwd
- B. ls
- C. cd
- D. mkdir

Answer: B

2. Which command displays the current working directory?

- A. ls
- B. cd
- C. pwd
- D. dir

Answer: C

3. Which command is used to create a new directory named college?

- A. create college
- B. mkdir college
- C. md college
- D. new college

Answer: B

4. Which command changes the current directory to college?

- A. move college
- B. cd college
- C. goto college
- D. open college

Answer: B

5. Which command creates an empty file named notes.txt?

- A. make notes.txt
- B. touch notes.txt
- C. file notes.txt
- D. create notes.txt

Answer: B

Scenario 2: Viewing and Managing Files

A user has created a file called report.txt and wants to examine and manage it.

1. Which command displays the contents of report.txt?

- A. cat report.txt
- B. show report.txt
- C. viewfile report.txt
- D. display report.txt

Answer: A

2. Which command copies report.txt to backup.txt?

- A. mv report.txt backup.txt
- B. cp report.txt backup.txt
- C. copy report.txt backup.txt
- D. duplicate report.txt backup.txt

Answer: B

3. Which command renames report.txt as final.txt?

- A. cp report.txt final.txt
- B. mv report.txt final.txt
- C. rn report.txt final.txt
- D. renamefile report.txt final.txt

Answer: B

4. Which command removes a file named old.txt?

- A. delete old.txt
- B. erase old.txt
- C. rm old.txt
- D. removefile old.txt

Answer: C

5. Which command provides detailed information about files, including permissions and ownership?

- A. ls -l
- B. pwd -l
- C. cd -l
- D. file -l

Answer: A

Thank You!

